

60016 OPERATIONS RESEARCH

Introduction to Operations Research

Lecture 01

Course Information

- ▶ Lecturers:
 - ▶ Dario Paccagnan, Ruth Misener
- ▶ Teaching Assistants
- ▶ Logistics for the first half (DP W1-W5 included):
 - ▶ Lectures/Tutorials: Mon 14:00-16:00 Huxley 311, Fri 14:00-16:00 Huxley 308
 - ▶ Detailed schedule (tentative) available online $\implies$ Scientia
- ▶ Resources:
 - ▶ Lecture material $\implies$ Scientia
 - ▶ Videos $\implies$ Panopto
 - ▶ Course forum $\implies$ Edstem
- ▶ One [assessed coursework](#), due Nov 19 (20% of final mark)

Some Books (Optional Readings)

- ▶ *F.Hillier & J.Lieberman:*
Introduction to Operations Research
- ▶ *H.Taha:*
Operations Research
- ▶ *D.G.Luenberger & Y.Ye:*
Linear and Nonlinear Programming
- ▶ *J.G.Eckert & M.Kupferschmid:*
Introduction to Operations Research

What is Operations Research?

- ▶ OR is a **multidisciplinary branch of mathematics** involving
 - ▶ mathematical modelling
 - ▶ mathematical optimisation
 - ▶ statistical analysis

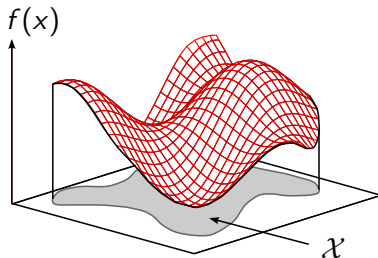
in order to find **"good" solutions** for **complex decision problems**.

- ▶ Typical **objectives** in OR are:
 - ▶ maximise **profit**
 - ▶ minimise **cost**
 - ▶ minimise **risk**
 - ▶ minimise **completion time**
 - ▶ maximise **efficiency**
 - ▶ etc.

Mathematical Programming

OR solves **mathematical programming models**:

$$\begin{array}{ll}\text{minimise} & z = f(x) \\ \text{subject to} & x \in \mathcal{X},\end{array}$$



where

- ▶ $x \in \mathbb{R}^n$ are the **decision variables**
- ▶ $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the **objective function** (e.g., cost)
- ▶ $\mathcal{X} \subseteq \mathbb{R}^n$ is the **feasible set** (set of admissible decisions)
- ▶ any vector x that minimises f is an **optimal solution** of the program and is denoted by x^*
- ▶ $z^* = f(x^*)$ is the **optimal value** achieved by x^*

Scope of OR

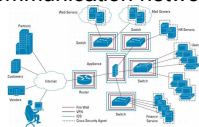
OR techniques are ubiquitous in **operations management**, **industrial engineering**, **economics and finance**, **ICT management**, **machine learning**, **health care**, **security**, etc.

Designing the layout of a factory

road traffic management



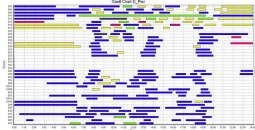
Constructing a telecommunication network



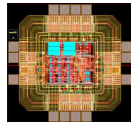
Determining the routes of city buses



scheduling project tasks



Designing the layout of a computer chip



Financial planning

History of OR

19th century:
Industrial Revolution



Efficiency of
production processes

World War II:
Birth of Modern OR



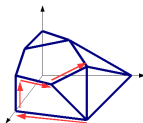
Optimal design of
convoy system



Where to add armour
in RAF bombers?

Allocate scarce resources to
various military operations
in an effective manner

1947
Simplex Algorithm



1953
Dynamic Programming



1970's
Personal Computers



Industry-size problems
can be solved efficiently

Phases of an OR Study

Hillier and Lieberman p. 8:

1. Define the problem of interest and **gather relevant data**.
2. Formulate a **mathematical model** to represent the problem.
3. Develop a **computer-based procedure** for deriving solutions to the problem from the model.
4. **Test** the model and **refine** as needed.
5. **Prepare for** the ongoing **application** of the model as prescribed by management.
6. **Implement**.

Course focuses on phases 2 and 3.

Topics of this Course

- ▶ Linear Programming
- ▶ Duality
 - ▶ CW: Game theory (application of linear optimisation & duality)
- ▶ Integer Programming
- ▶ Optimization Modelling

Extra (optional): GLPK case studies

Linear Programming

- ▶ A **Linear program (LP)** is a mathematical program that
 - ▶ optimises (maximises or minimises) a **linear objective function**
 - ▶ over a **feasible set** described by **linear** equality and/or inequality constraints.
- ▶ Optimal decision tool
- ▶ Widely adopted, many success stories (see Hillier & Lieberman)
- ▶ This course addresses linear optimisation
 - ▶ *70007 - Computational Optimisation* deals with the theory of non-linear optimisation.

Mentimeter 1: is this an LP?

$$\begin{array}{ll}\text{maximise} & z = x_1 + x_2 \\ \text{subject to} & x_1 + 3x_2 \leq 3 \\ & x_1 = 3 \\ & x_1, x_2 \geq 0\end{array}$$

Mentimeter 2: is this an LP?

$$\begin{array}{ll}\text{maximise} & z = x_1 + x_2 \\ \text{subject to} & (x_1 + 3x_2)^2 \leq 3 \\ & x_1 = 3 \\ & x_1, x_2 \geq 0\end{array}$$

Mentimeter 3: is this an LP?

$$\begin{array}{ll}\text{maximise} & z = x_1 + x_2 \\ \text{subject to} & x_1 + x_1 x_2 \leq 3 \\ & x_1 = 3 \\ & x_1, x_2 \geq 0\end{array}$$

Note: same feasible set as in Mentimeter 1, since $x_1 = 3$

LP Running Case: Example 1

Resource Allocation Problem: A manufacturer produces A (acid) and C (caustic) and wants to decide a production plan.

Ingredients used for producing A and C are: X (e.g., a sulphate) and Y (e.g., sodium).

- ▶ Each ton of A requires: 2ton of X; 1ton of Y
- ▶ Each ton of C requires: 1ton of X ; 3ton of Y
- ▶ Supply of X limited to: 11ton/week
- ▶ Supply of Y limited to: 18ton/week
- ▶ A sells for: £1000/ton
- ▶ C sells for: £1000/ton
- ▶ Market research: max 4 tons of A/week can be sold.

Maximize weekly value of sales of A and C.
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Example 1 (Modelling)

How much A and C to produce?

⇒ Formulate a **mathematical programming model**!

- ▶ **Decision variables**

- ▶ x_1 = weekly production of A (in tons)
- ▶ x_2 = weekly production of C (in tons)

- ▶ **Objective function**

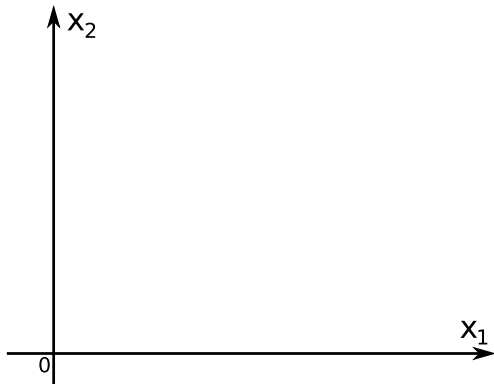
- ▶ $z = f(x_1, x_2)$ = weekly profit (in 1000 £)

- ▶ **Feasible set**

- ▶ $\mathcal{X}$ = set of all implementable/admissible production plans
 $x = (x_1, x_2)$
- ▶ e.g., $x = (27, 2)$ is not possible (not enough supply!)

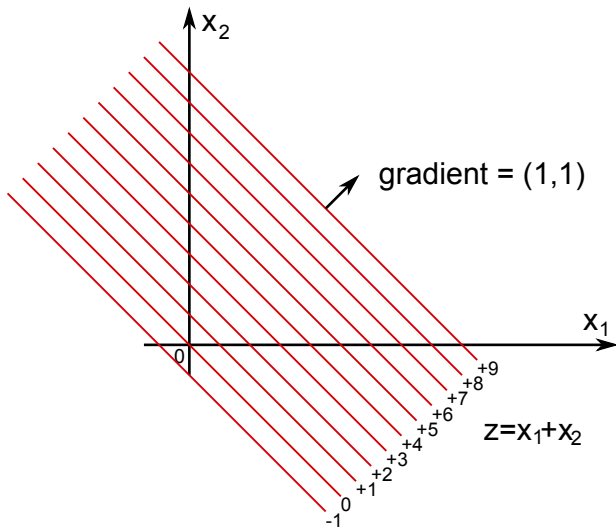
Example 1 (Decision Variables)

A **production plan** is representable as $x = (x_1, x_2)$



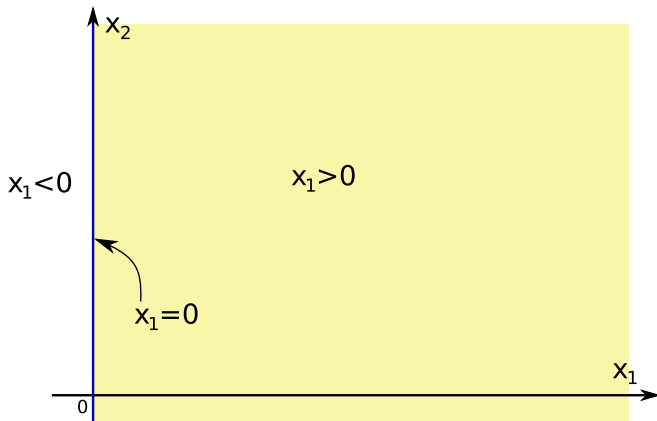
Example 1 (Objective Function)

Profit: $z = x_1 + x_2$ (in 1000 £)



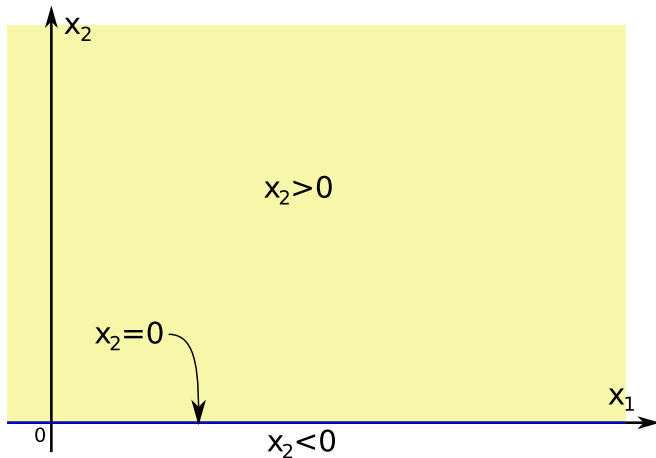
Example 1 (Feasible Set)

Amount of A produced is **non-negative**: $x_1 \geq 0$



Example 1 (Feasible Set)

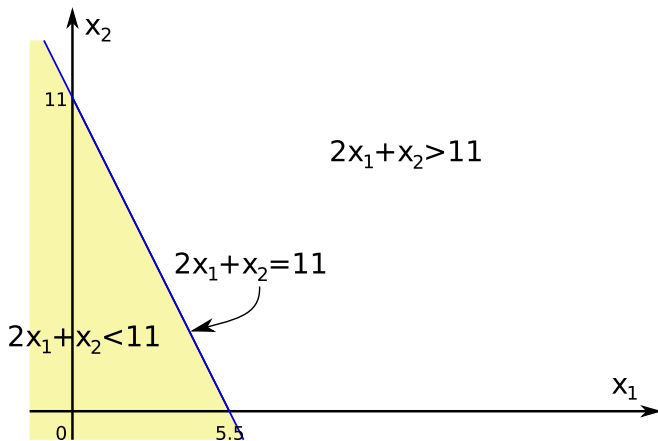
Amount of C produced is **non-negative**: $x_2 \geq 0$



Example 1 (Feasible Set)

x_1 tons of A & x_2 tons of C require $2x_1 + x_2$ ton of X

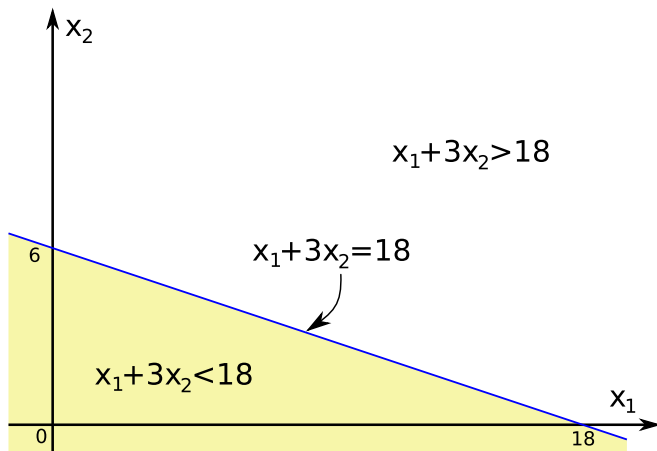
X is limited to 11ton/week: $2x_1 + x_2 \leq 11$



Example 1 (Feasible Set)

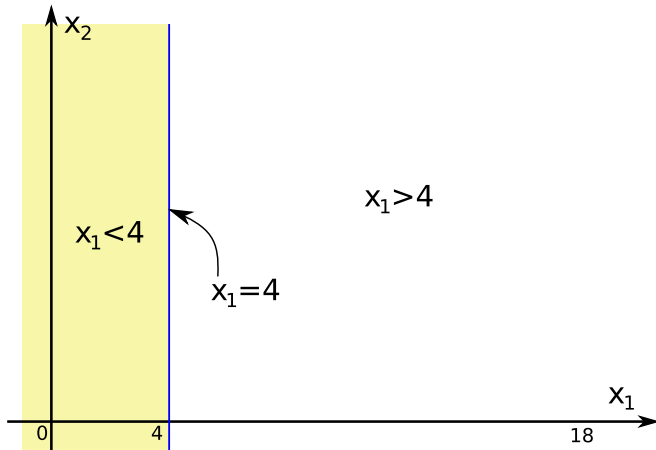
x_1 tons of A & x_2 tons of C require $x_1 + 3x_2$ ton of Y

Y is limited to 18ton/week: $x_1 + 3x_2 \leq 18$



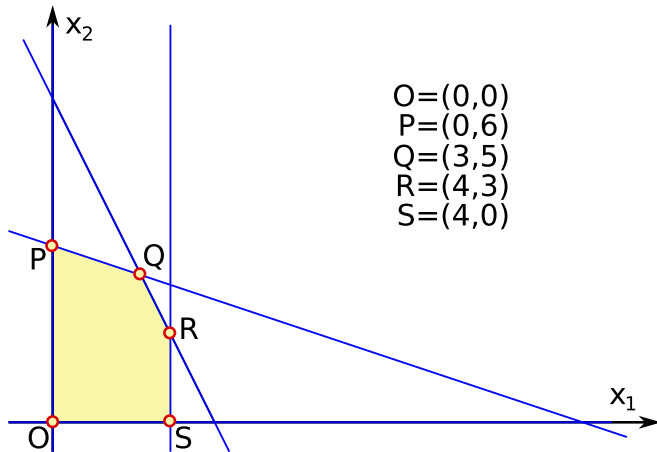
Example 1 (Feasible Set)

Cannot sell more than 4 tons of A/week: $x_1 \leq 4$



Example 1 (Feasible Set)

To obtain the overall feasible set,
intersect the feasible sets of all individual constraints



Example 1 (Feasible Set)

- ▶ The feasible set is a **convex polygon**
- ▶ The corner points O,P,Q,R,S of the feasible set are termed **vertices**
- ▶ Each vertex is given by the **intersection of two blue lines**
 - ▶ its coordinates can be computed by jointly solving the two linear equations defining the blue lines
- ▶ We obtain $O=(0,0)$, $P=(0,6)$, $Q=(3,5)$, $R=(4,3)$, $S=(4,0)$

Example 1 (Summary)

The **best production plan** is obtained by solving the following mathematical problem:

maximise $z = x_1 + x_2$: **objective function**

subject to $2x_1 + x_2 \leq 11$: **constraint on availability of X**

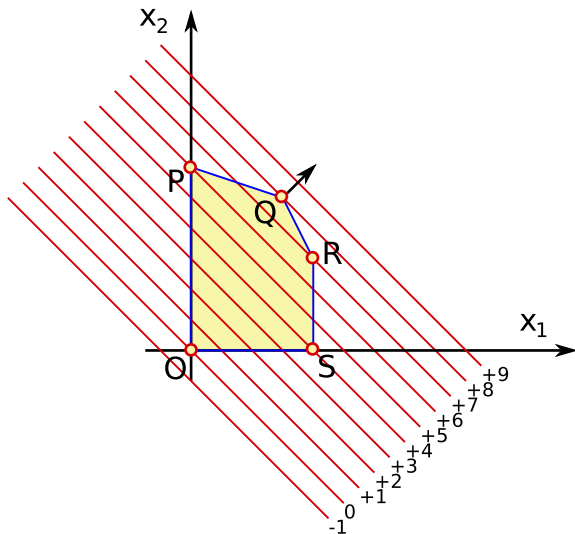
$x_1 + 3x_2 \leq 18$: **constraint on availability of Y**

$x_1 \leq 4$: **constraint on demand of A**

$x_1, x_2 \geq 0$: **non-negativity constraints**

This is a **linear program**.

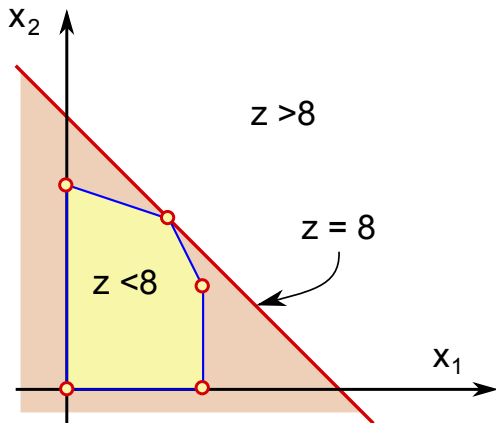
Example 1 (Graphical Solution)



Example 1 (Graphical Solution)

All feasible points satisfy $z \leq 8$

Q is the **only feasible point** (x_1, x_2) with $z = x_1 + x_2 = 8$



Linear Programming

- ▶ Assume feasible set $\mathcal{X}$ bounded and nonempty
- ▶ We can prove that LPs have an **optimal vertex** solution
 - ▶ LPs may be solved by inspecting all vertices, but ...
 - ▶ The number of vertices grows **exponentially** with the number of constraints and variables in the LP
- ▶ How to program a computer to **efficiently** solve LPs?
 - ▶ **Simplex Algorithm** finds an optimal vertex
 - ▶ Vertices inspected by the Simplex algorithm are often a small subset of the total
 - ▶ What made the Simplex algorithm famous is that it works well on most instances, i.e., an excellent average behaviour

Variants of Example 1

- ▶ Minimise $z = 3x_1 - x_2$ over feasible set of Example 1

Examine value of objective function at all vertices:

O=(0,0)	P=(0,6)	Q=(3,5)	R=(4,3)	S=(4,0)
0	-6	4	9	12

$\Rightarrow$ P: $x_1 = 0$, $x_2 = 6$ is optimal.

- ▶ Maximise $z = 2x_1 + x_2$ over feasible set of Example 1:

Any point on the line segment QR is optimal.

$\Rightarrow$ points other than vertices can be optimal, but there is at least one optimal vertex

Variants of Example 1

- ▶ Set a minimum weekly production goal of 7 tons of C

We add a new constraint $x_1 \geq 7$. The the feasible set becomes **empty**, because we previously imposed $x_1 \leq 4$

⇒ the LP is infeasible

- ▶ Remove constraints on availability of X and Y

Objective function can now grow to $+\infty$ on the feasible set.
There is no maximum!

⇒ the LP is unbounded

Diet Problem

Determine **most economical diet**, with **basic nutritional requirements** for good health.

- ▶ n different foods: i th sells at price c_i /unit,
- ▶ m basic **nutritional ingredients**: j th ingredient's daily intake for individual is at least b_j units (healthy diet),
- ▶ one unit of food i contains a_{ji} units of the j th ingredient,
- ▶ x_i : units of food i in diet (we allow fractional amounts).

Nutrition Facts	
Serving Size 1 package (456g)	
Amount Per Serving	Calories from Fat 570
Calories 1,630	
% Daily Value*	
Total Fat 64g	98%
Saturated Fat 21g	104%
Cholesterol 690mg	231%
Sodium 2,090mg	87%
Total Carbohydrate 78g	26%
Dietary Fiber 4g	17%
Sugars 22g	
Protein 36g	
Vitamin A 25%	Vitamin C 2%
Calcium 20%	Iron 25%
* Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs:	
	Calories: 2,000 2,500
Total Fat	Less than 65g 80g
Sat Fat	Less than 20g 25g
Cholesterol	Less than 300mg 300mg
Sodium	Less than 2,400mg 2,400mg
Total Carbohydrate	300g 375g
Dietary Fiber	25g 30g

Diet Problem (cont)

minimise total cost:

$$z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

subject to:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & \geq & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & \geq & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & \geq & b_m \end{array}$$

non-negativity of food quantities

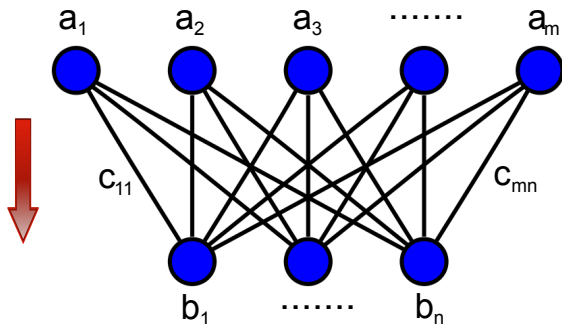
$$x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$$

Transportation Problem

- ▶ Quantities $a_1, a_2, \dots, a_m$ of a product to be shipped from m locations,
- ▶ demanded in amounts $b_1, b_2, \dots, b_n$ at n destinations,
- ▶ c_{ij} : unit cost of transporting product from i to j ,
- ▶ x_{ij} : amounts to be shipped from i to j ($i = 1, \dots, m$; $j = 1, \dots, n$). Assume to allow fractional amounts.

Determine x_{ij} to satisfy shipping requirements and minimise total cost.

Transportation Problem (cont)



Transportation Problem (cont)

minimise total cost: $z = \sum_{i,j} c_{ij} x_{ij}$

subject to:

$$\sum_{j=1}^n x_{ij} = a_i \quad (\text{total shipped from } i = 1, \dots, m)$$

$$\sum_{i=1}^m x_{ij} = b_j \quad (\text{total required by } j = 1, \dots, m)$$

$$x_{ij} \geq 0 \quad \text{consistency: } \sum_{i=1}^m a_i = \sum_{j=1}^n b_j$$

Air Traffic Control

- ▶ Airplane n , $n = 1, \dots, N$, arrives at the airport within the time interval $[a_n, b_n]$.
- ▶ For safety reasons, the airport wants to set the arrival time for each plane such that the **time interval between two successive flights is as large as possible**. Planes land in sequential order: $1, 2, \dots, N$.
- ▶ Denote by t_n be the arrival time of airplane n , and let Δ be the smallest inter-arrival time.

$$\text{maximise} \quad z = \Delta$$

subject to:

$$t_n - t_{n-1} - \Delta \geq 0 \quad n = 2, \dots, N$$

$$a_n \leq t_n \leq b_n \quad n = 1, \dots, N$$

Newsboy Problem

- ▶ In the morning, the newsboy buys x newspapers from the publisher at a wholesale price c per piece.
- ▶ The publisher provides at most u newspapers.
- ▶ The demand (amount of newspapers that can be sold) is uncertain. It amounts to d_s with probability $p_s > 0$ for different scenarios $s = 1, \dots, S$, where $\sum_{s=1}^S p_s = 1$. The d_s and p_s values are given.
- ▶ During the day, the newsboy sells y_s newspapers in scenario s at retail price $q > c$.
- ▶ In the evening, unsold newspapers become worthless.
- ▶ What is the newsboy's strategy to **maximise average profit**?

Newsboy Problem (cont)

maximise average profit:

$$z = -c x + q \sum_{s=1}^S p_s y_s$$

subject to:

$$0 \leq x \leq u \quad (\text{amount bought in the morning})$$

$$0 \leq y_s \leq x \quad (\text{sell at most } x \text{ in each scenario } s = 1, \dots, S)$$

$$y_s \leq d_s \quad (\text{demand constraint for } s = 1, \dots, S)$$

Data Fitting

- ▶ Given input variables $\vec{a}_n \in \mathbb{R}^d$ and output variables b_n , $n = 1, \dots, N$; e.g. $\vec{a}_n = (\text{height}, \text{age})$ and $b_n = \text{weight of person } n$.
- ▶ Is there a linear relationship between inputs and outputs?
- ▶ Equivalently, is there an $\vec{x} \in \mathbb{R}^d$ with the property that $\vec{a}_n^T \vec{x} \approx b_n$?
- ▶ The **least squares problem** searches for a $\vec{x}$ such that

$$z = \sum_{n=1}^N (\vec{a}_n^T \vec{x} - b_n)^2$$

is minimised.

Data Fitting (cont)

- ▶ To give **less/more weight to outliers**, one can choose different objective criteria. We can minimise

$$z = \sum_{n=1}^N |\vec{a}_n^T \vec{x} - b_n| \quad \text{or} \quad z = \max_n |\vec{a}_n^T \vec{x} - b_n|.$$

- ▶ We will see that these problems can be rewritten as **linear programs**.